

$|k_m\rangle \forall m < n$ implies that the support of $|k_n\rangle$ folds back on the Fock-space graph, and thus in principle exists on the entire graph. This is shown schematically in Fig. 1 for the case of $L = 6$. Indeed, since the Krylov chain has a length $N_{\mathcal{H}} = 2^L$, a fraction tending to unity of the sites thereon potentially have support over the entire Fock space.

It is of course important to establish that the Krylov-basis states indeed span the Fock-space graph, whether in the ergodic or MBL regimes. To this end, we define an effective size of the wave function corresponding to the Krylov orbital $|k_n\rangle$ on the Fock-space graph, via

$$\mathcal{R}_n = \sum_{r=0}^L r \binom{L}{r} w_n(r) = \sum_{r=0}^L r \sum_{I: r_{II_0}=r} w_{nI}. \quad (14)$$

Physically, $\mathcal{R}_n$ gives the mean Hamming distance on the Fock-space graph that is associated with the given Krylov $|k_n\rangle$, and as such is a bona fide measure of how the Krylov-basis state is spread over the Fock-space graph. Since $\{|k_n\rangle\}$ forms a complete orthonormal basis, such that $\sum_n w_n(r) = 1$ (Eq. 13), note that $\mathcal{R}_n$ satisfies a sum rule

$$\sum_{n=0}^{N_{\mathcal{H}}-1} \mathcal{R}_n = \sum_{r=0}^L r \binom{L}{r} = N_{\mathcal{H}} L / 2. \quad (15)$$

Sufficiently deep in the ergodic phase, one expects $w_{nI} \sim N_{\mathcal{H}}^{-1}$ and hence $w_n(r) \sim N_{\mathcal{H}}^{-1}$ for all I and $n > L$; which when used in Eq. 14 yields $\mathcal{R}_n \simeq L/2$. That this is indeed the case is confirmed by numerical results shown in Fig. 2(a) for a disorder strength $W = 1$ deep in the ergodic phase (and with system sizes L as indicated).

The natural generalisation of this to encompass the MBL regime is the binomial form

$$w_n(r) = p_n^r (1 - p_n)^{L-r}, \quad (16)$$

which from Eq. 14 yields $\mathcal{R}_n = p_n L$ (with the deep ergodic phase corresponding to $p_n = 1/2$). The sum rule in Eq. 15 then implies $\sum_n p_n = N_{\mathcal{H}}/2$. Assuming $p_n = g(n/N_{\mathcal{H}}^a)$ to be some function of $n/N_{\mathcal{H}}^a$ (with exponent a), this sum rule can be cast for $N_{\mathcal{H}} \gg 1$ as

$$\int_0^{N_{\mathcal{H}}} dn g\left(\frac{n}{N_{\mathcal{H}}^a}\right) = N_{\mathcal{H}}^a \int_0^{N_{\mathcal{H}}^{1-a}} dx g(x) = \frac{N_{\mathcal{H}}}{2}, \quad (17)$$

which naturally requires $a = 1$. This then implies that $\mathcal{R}_n$ satisfies the scaling form

$$\mathcal{R}_n = L \times g\left(\frac{n}{N_{\mathcal{H}}}\right), \quad (18)$$

such that $\mathcal{R}_n/L$ is a function solely of $n/N_{\mathcal{H}}$. While the scaling form is quite trivially satisfied in the ergodic regime (as above), numerical results in the MBL phase, shown in Fig. 2(b) for disorder strength $W = 10$, confirm this to be case in general. The key upshot of this scaling

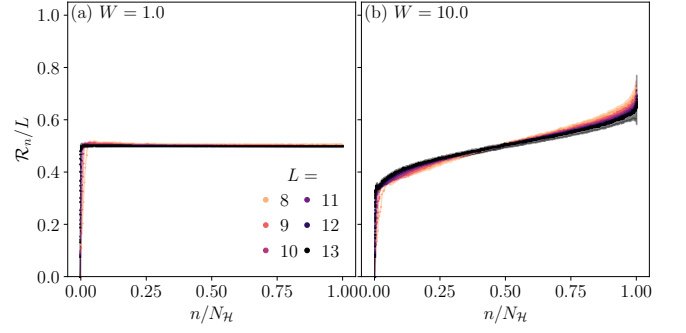


FIG. 2. Numerical results for the effective size of the wave function on the Fock-space graph corresponding to the n^{th} Krylov orbital, as quantified by $\mathcal{R}_n$ defined in Eq. 14 (with system sizes L indicated). Panels (a) and (b) with $W = 1$ and $W = 10$ exemplify, respectively, the ergodic and MBL regimes. The results conform to the scaling form in Eq. 18. The (thin) shaded region around the data points reflects the statistical errors over disorder realisations.

form is that in both the ergodic and MBL regimes, the Krylov-basis states indeed have access to the entire Fock-space graph¹.

IV. SCALING FORMS FOR SPREAD COMPLEXITY AND EIGENSTATE AMPLITUDES

With the basic properties of the Krylov chain and the spread complexity defined, we turn now to the Krylov-space anatomy of the eigenstates.

A. Distribution and scaling of $S_{K,\infty}$

We start with the distribution of the infinite-time spread complexity over disorder realisations; in particular the distribution $P_s(s)$ of the late-time complexity rescaled by its mean, $s = S_{K,\infty}/\langle S_{K,\infty} \rangle$, numerical results for which are shown in Fig. 3 for representative W 's in each of the ergodic and MBL phases.

In the ergodic regime the distributions are well described by a Gaussian with mean 1

$$P_s(s) = \frac{1}{\sqrt{2\pi\sigma_s^2}} \exp\left[-\frac{(s-1)^2}{2\sigma_s^2}\right], \quad (19)$$

and clearly narrow rapidly with increasing system size. Results for the mean $\langle S_{K,\infty} \rangle$ are given in Fig. 4(a), from which it is evident that in the ergodic regime,

$$\langle S_{K,\infty} \rangle \simeq N_{\mathcal{H}}/2. \quad (20)$$

¹ This is not of course the case for the first L Krylov basis states, but that is immaterial since these constitute an exponentially small (in L) fraction of the 2^L Krylov states.